LECTURE NOTES

From the PCP Theorem to Unique Games

Proofs, algorithms and geometry
of hardness of approximation

Sixteen lectures from *proof checking* to the *Unique Games Conjecture*
Draft of October 10, 2026

About these notes

These notes retrace one of the most fruitful lines of research in theoretical computer science: how the question *"is there a proof of this statement?"* turned into the question *"how well can this optimization problem be approximated?"*, and how the second question, in turn, led to a single conjecture—the *Unique Games Conjecture* (UGC)—that seems to explain the answer for a startlingly large family of problems. The plan is one lecture per paper-sized idea: the language of games and the PCP theorem (Lectures 1–4), amplification by parallel repetition and by Dinur's gap amplification (5–6), Håstad's optimal inapproximability (7), the UGC and its consequences in analysis, geometry and semidefinite programming (8–12), the 2-to-2 Games theorem (13–15), and a critical reading of the 2026 claim of a full proof of the UGC (16).

Prerequisites. Undergraduate algorithms and the basics of NP-completeness. Everything else (approximation ratios, gap problems, games, ...) is defined when first needed.

How to read the boxes.

- **Blue boxes** summarize the big picture or a key idea; if you are short on time, read the blue boxes and the statements of the theorems.
- **Orange boxes** flag common misconceptions.
- **Wine-red boxes** state the *epistemic status* of a result. Whenever a claim has not been independently verified, we say so there and nowhere do we treat it as established.

How results are labelled. A result with an attribution in its heading, e.g. "Theorem (PCP theorem; Arora–Safra, Arora–Lund–Motwani–Sudan–Szegedy)", is quoted from the literature; we give a proof only if one follows the statement, and otherwise point to where the proof can be found. Statements that depend on an unproven conjecture carry that conjecture in their name, e.g. "Theorem (assuming the UGC)".

Exercises. Each lecture ends with exercises marked [$\star$] (routine), [$\star\star$] (needs an idea) or [$\star\star\star$] (substantial). Several exercises come with a hint.

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LECTURE 1

Why PCP and Unique Games Matter

From checking proofs to approximating optima

Color every vertex of a graph red or blue, and call an edge *cut* if its endpoints get different colors. Flipping a fair coin for each vertex cuts every edge with probability $\frac{1}{2}$, so some coloring cuts at least half of all edges. In 1994 Goemans and Williamson (Goemans and Williamson, 1995) showed how to do much better: using semidefinite programming and a random hyperplane, one can find in polynomial time a cut that contains at least $0.8785\ldots$ times as many edges as a *maximum* cut. More than thirty years later that constant has not moved. Is it a limit of our imagination, or a limit of mathematics?

Here is a remarkable partial answer. The best *unconditional* hardness result says that, unless $\mathsf{P} = \mathsf{NP}$, no polynomial-time algorithm beats $16/17 \approx 0.941$. But if the *Unique Games Conjecture* (UGC) is true, then no polynomial-time algorithm can beat $0.8785\ldots + \varepsilon$ for any $\varepsilon > 0$ (Khot et al., 2007; Mossel et al., 2010): the strange constant defined by a calculus minimization is *exactly* the truth. The conjecture was formulated by Khot in 2002 (Khot, 2002). As we write this (October 2026) it is in an unusual position: a manuscript dated 23 September 2026 claims a proof (Section 1.8). We treat the conjecture as a conjecture throughout, and say so whenever a result depends on it.

Every step of this story—what "hard to approximate" means, why we can prove it at all, and why the conjecture takes such a peculiar form—traces back to a single idea:

> *To prove that an optimization problem is hard to approximate, we need proofs that can be checked by reading only a few bits.*

This lecture builds the vocabulary (gap problems, probabilistically checkable proofs, two-prover games, Label Cover, Unique Games), proves the handful of short results that explain *why* the vocabulary looks the way it does, and draws the map on which the remaining fifteen lectures sit.

After this lecture you should be able to…

(1) define a gap problem, and explain how NP-hardness of a gap problem rules out approximation algorithms (Lemma 1.4);
(2) explain why the classical reductions of the 1970s cannot produce a gap, and what the PCP theorem adds (Section 1.4);
(3) translate between PCP verifiers, CSPs, two-prover games and Label Cover, and prove the two forms of the PCP theorem equivalent (Proposition 1.9);
(4) state the Unique Games Conjecture with its quantifiers in the right order, and say why a Unique Game with perfect completeness is easy;
(5) locate each later lecture on a map of the field (Section 1.8).

Problem	Best algorithm	NP-hard beyond	If the UGC holds, hard beyond
Max-E3-Lin($\mathbb{F}_2$)	$\frac{1}{2}$ (random)	$\frac{1}{2}+\varepsilon$ (Håstad, 2001)	(already tight)
Max-E3-SAT	$\frac{7}{8}$ (random)	$\frac{7}{8}+\varepsilon$ (Håstad, 2001)	(already tight)
Max-Cut	$\alpha_{\mathrm{GW}} \approx 0.8786$ (Goemans and Williamson, 1995)	$\frac{16}{17}+\varepsilon \approx 0.941$ (Håstad, 2001; Trevisan et al., 2000)	$\alpha_{\mathrm{GW}}+\varepsilon$ (Khot et al., 2007; Mossel et al., 2010)
Vertex Cover	2 (maximal matching)	$\sqrt{2}-\varepsilon \approx 1.414$ (Dinur et al., 2018; Khot et al., 2018)	$2-\varepsilon$ (Khot and Regev, 2008)

Table 1.1: Approximability thresholds for four basic problems. Ratios are ≤ 1 for maximization and ≥ 1 for minimization problems (Definition 1.1). "NP-hard beyond α" means that no polynomial-time algorithm achieves ratio better than α unless P = NP. The first two rows are unconditionally tight; for the last two, the UGC would close the gap between algorithm and hardness.

1.1 A scoreboard of approximability

Let us begin with the answer key. Table 1.1 lists four problems, the best approximation ratio known for each, and what is known about the *limit* of approximability.

The first two rows are the great success of the PCP method, and they are astonishing. Max-E3-Lin($\mathbb{F}_2$) asks for an assignment satisfying as many as possible of a given list of linear equations $x_i + x_j + x_k = b \pmod 2$. A uniformly random assignment satisfies half of the equations, whatever the system is. Håstad proved that nothing polynomial-time does better, even if you are promised that a $(1-\varepsilon)$-fraction of the equations can be satisfied simultaneously: *guessing is optimal*. The same is true for Max-E3-SAT, where a random assignment satisfies a $\frac{7}{8}$-fraction of the clauses.

The last two rows are where the story stops being a triumph. For Max-Cut and Vertex Cover the best algorithms (a clever semidefinite program; a trivial maximal matching) leave a visible gap to the best hardness proofs. Assuming the UGC, both gaps close *simultaneously* and *exactly*. In Lecture 11 we shall see that this is not a coincidence: Raghavendra (Raghavendra, 2008) proved that, assuming the UGC, for *every* constraint satisfaction problem the optimal approximation ratio is the integrality gap of one generic semidefinite program. In the other direction, the constant in the Max-Cut row is

$$\alpha_{\mathrm{GW}} \;=\; \min_{0<\theta\leq\pi} \; \frac{2}{\pi}\cdot\frac{\theta}{1-\cos\theta} \;\approx\; 0.87856, \tag{1.1}$$

and the hardness proof of Khot, Kindler, Mossel and O'Donnell reproduces precisely this minimization, with θ the angle between two random-hyperplane vectors. We will see in Lecture 9 how the same expression emerges from a Fourier-analytic argument about Gaussian noise.

The question behind the whole course

For which problems can an efficient algorithm do better than the trivial one, by how much, and how do we *prove* that nothing does better? The PCP theorem answers this for problems like Max-E3-SAT; the Unique Games Conjecture (still unproven) would answer it for a much larger class.

1.2 Approximation algorithms in one page

All problems in this lecture are constraint satisfaction problems or close relatives. An *instance* of a Max-CSP consists of variables $x_1, \dots, x_n$ with values in a finite domain and a list of m *constraints*, each depending on a few variables. The *value* of an assignment is the *fraction* of constraints it satisfies, and $\mathrm{OPT}(I)$ is the maximum value over all assignments. Examples: Max-3SAT (clauses with at most three literals), Max-E3-Lin($\mathbb{F}_2$) (equations $x_i + x_j + x_k = b$), Max-Cut (constraints $x_u \neq x_v$, one per edge).

Definition 1.1 (Approximation ratio)**.** A polynomial-time algorithm $\mathcal{A}$ is an α-*approximation* for a maximization problem ($0 < \alpha \leq 1$) if on every instance I it outputs a solution of value at least $\alpha \cdot \mathrm{OPT}(I)$. For a minimization problem $\mathcal{A}$ is a ρ-*approximation* ($\rho \geq 1$) if it outputs a solution of value at most $\rho \cdot \mathrm{OPT}(I)$. The *approximation threshold* of a problem is the best constant α (or ρ) for which an approximation algorithm exists.

Example 1.2 (Four one-paragraph algorithms)**.**

- Max-Cut, *ratio* $\frac{1}{2}$. Put each vertex on a random side. Every edge is cut with probability $\frac{1}{2}$, so the expected number of cut edges is $m/2 \geq \mathrm{OPT}/2$.
- Max-E3-Lin($\mathbb{F}_2$), *ratio* $\frac{1}{2}$. For a random assignment and any fixed equation $x_i + x_j + x_k = b$, condition on x_i, x_j: exactly one value of x_k satisfies the equation, so it holds with probability $\frac{1}{2}$.
- Max-E3-SAT, *ratio* $\frac{7}{8}$ (every clause has three literals on three distinct variables). A clause is violated by exactly one of the eight settings of its variables, hence is satisfied with probability $\frac{7}{8}$.
- Vertex Cover, *ratio* 2. Let M be a maximal matching. Its $2|M|$ endpoints form a vertex cover (an uncovered edge could be added to M), while any cover must contain one endpoint of each of the $|M|$ disjoint edges of M, so $\mathrm{OPT} \geq |M|$.

The three randomized algorithms can be derandomized by the method of conditional expectations (Exercise 1.1).

So an algorithm that does *nothing intelligent* already achieves a constant ratio. The whole subject starts from the question whether intelligence can ever pay off—and for Max-E3-SAT and Max-E3-Lin it turns out that it cannot. The tool that proves this is the subject of the next three sections.

1.3 Gap problems: the language of hardness

How does one prove that an *approximation* problem is NP-hard? Decision problems are what the theory of NP-completeness knows how to handle, so we turn the approximation problem into a decision problem—a *promise* problem.

Definition 1.3 (Gap problem)**.** Let Π be a Max-CSP and $0 \le s < c \le 1$. The *gap problem* $\textsc{Gap}_{c,s}$-Π is the following promise problem. Given an instance I of Π such that either

$$\text{(YES)}\quad \mathrm{OPT}(I) \ge c \qquad \text{or} \qquad \text{(NO)}\quad \mathrm{OPT}(I) \le s,$$

decide which of the two cases holds. On instances outside both cases, any answer is acceptable. We call c the *completeness* and s the *soundness*; the gap problem has *perfect completeness* if $c = 1$. The problem is NP*-hard* if there is a polynomial-time map f from 3-CNF formulas to instances of Π with φ satisfiable $\Rightarrow$ $f(\varphi)$ is a YES instance, and φ unsatisfiable $\Rightarrow$ $f(\varphi)$ is a NO instance.

Lemma 1.4 (Gap hardness implies inapproximability)**.** *If* $\textsc{Gap}_{c,s}$-Π *is* NP*-hard, then for every* $\alpha > s/c$ *there is no polynomial-time* α*-approximation algorithm for* Π *unless* $\mathsf{P} = \mathsf{NP}$.

Proof. Suppose $\mathcal{A}$ is a polynomial-time α-approximation with $\alpha > s/c$. We use it to decide 3SAT. Given φ, compute $I = f(\varphi)$, run $\mathcal{A}$ on I, evaluate the value v of the solution it returns (this is easy), and accept iff $v > s$. If φ is satisfiable then $\mathrm{OPT}(I) \ge c$, so $v \ge \alpha c > s$. If φ is unsatisfiable then $v \le \mathrm{OPT}(I) \le s$. Hence this is a polynomial-time algorithm for 3SAT. □

Conversely, an approximation algorithm with ratio below s/c would solve the gap problem—so the gap problem *is* the decision core of approximation. In this language, the rows of Table 1.1 read as follows:

Håstad: $\textsc{Gap}_{1-\varepsilon,\frac{1}{2}+\varepsilon}$-Max-E3-Lin and $\textsc{Gap}_{1,\frac{7}{8}+\varepsilon}$-Max-E3-SAT are NP-hard;

UGC: $\textsc{Gap}_{c,\alpha_{\mathrm{GW}}c+\varepsilon}$-Max-Cut is NP-hard (for suitable c).

Remark (Why completeness $1-\varepsilon$ rather than 1). Perfect completeness is not always attainable. A system of linear equations over $\mathbb{F}_2$ whose equations can *all* be satisfied is solved by Gaussian elimination, so $\textsc{Gap}_{1,s}$-Max-E3-Lin $\in \mathsf{P}$ for every $s < 1$ (Exercise 1.3). Håstad's theorem has to take $c = 1-\varepsilon$ for exactly this reason, and we will see the same phenomenon, for a similar reason, in the Unique Games Conjecture (Section 1.7).

1.4 Why ordinary reductions cannot create a gap

The NP-completeness theory of the 1970s (Cook, 1971; Levin, 1973; Karp, 1972) is a theory of *exact* problems, and its reductions are, in a precise sense, too gentle for approximation. This section explains why, and in doing so isolates what the PCP theorem must supply.

1.4.1 Cook–Levin yields a gap of one clause

Recall how 3SAT is shown NP-hard. Let $L \in \mathsf{NP}$ with verifier V and let x be an input. The Cook–Levin reduction outputs a formula φ_x that encodes the computation tableau of V on (x, w), where the witness w is left as free variables. The clauses say that (a) the first row of the tableau contains x, (b) every 2×3 window of the tableau is consistent with the transition function of V, and (c) the last row contains the accepting state.

Observation. For *every* x—also for $x \notin L$—there is an assignment to φ_x that violates only the clause in (c). Hence $\mathrm{OPT}(\varphi_x) \geq 1 - 1/m$, where m is the number of clauses.

Proof sketch. Choose any witness w and write down the true computation of V on (x, w) as a tableau. Constraints (a) and (b) hold by construction; (c) fails if V rejects (x, w), but it is a single clause. □

So the classical reduction maps YES instances to formulas with $\mathrm{OPT} = 1$ and NO instances to formulas with $\mathrm{OPT} = 1 - \frac{1}{m}$ exactly. It proves that $\textsc{Gap}_{1,1-1/m}$-3SAT is NP-hard, which is nothing but the statement that maximizing the number of satisfied clauses *exactly* is NP-hard. The gap is a single clause out of m, and by Lemma 1.4 it rules out only approximation ratios above $1 - \frac{1}{m}$, which is no restriction at all on an algorithm.

1.4.2 Karp reductions transport gaps, they do not create them

Suppose now that someone *gives us* a gap, and we want to carry it to another problem. The standard reduction from 3SAT to Independent Set does this perfectly.

Proposition 1.5 (The triangle reduction)**.** *Let φ be a 3-CNF with m clauses, and define the graph G_φ as follows: there is a vertex for every pair (clause C_j, literal $\ell \in C_j$); the three vertices of each clause form a triangle; and two vertices (j, ℓ), (j', ℓ') are adjacent when $\ell' = \neg\ell$. Then the size of a maximum independent set of G_φ equals the maximum number of clauses of φ that can be simultaneously satisfied.*

Proof. ($\geq$) Given an assignment satisfying t clauses, pick for each satisfied clause one satisfied literal. The t chosen vertices lie in different triangles, and no two of them carry complementary literals, since both are true under the same assignment. So they are independent. ($\leq$) Let S be an independent set. It contains at most one vertex per triangle, and the literals it contains are pairwise non-complementary, so they can all be set to true by a single assignment, which then satisfies at least $|S|$ clauses. □

Consequently $\textsc{Gap}_{c,s}$-3SAT reduces to the problem of distinguishing independent sets of size $\geq cm$ from independent sets of size $\leq sm$ in a graph with $3m$ vertices: the gap is carried over *unchanged*. The same is true for almost all the textbook reductions among NP-complete problems. They are *local gadget constructions* that preserve the quantity "how many constraints can be satisfied"; so they preserve a gap if there is one at the start, and they have none to preserve if there is none.

Gap creation versus gap preservation

Hardness of approximation has two ingredients. **(i) Gap creation**: one reduction from NP (e.g. from 3SAT) to a constraint satisfaction problem that turns "satisfiable" vs. "unsatisfiable" into "everything satisfiable" vs. "at most a $(1 - \varepsilon_0)$ fraction satisfiable." This is the PCP theorem (Section 1.5), and it does not follow from anything in the NP-completeness toolbox. **(ii) Gap preservation or amplification**: a long chain of reductions that carry the gap to other problems, sharpening its parameters on the way. Lectures 3, 4 and 6 are about (i); Lectures 5 and 7 are about (ii); and Lectures 8–16 explore what happens when ordinary gap amplification runs out of steam, which is where the Unique Games Conjecture comes in.

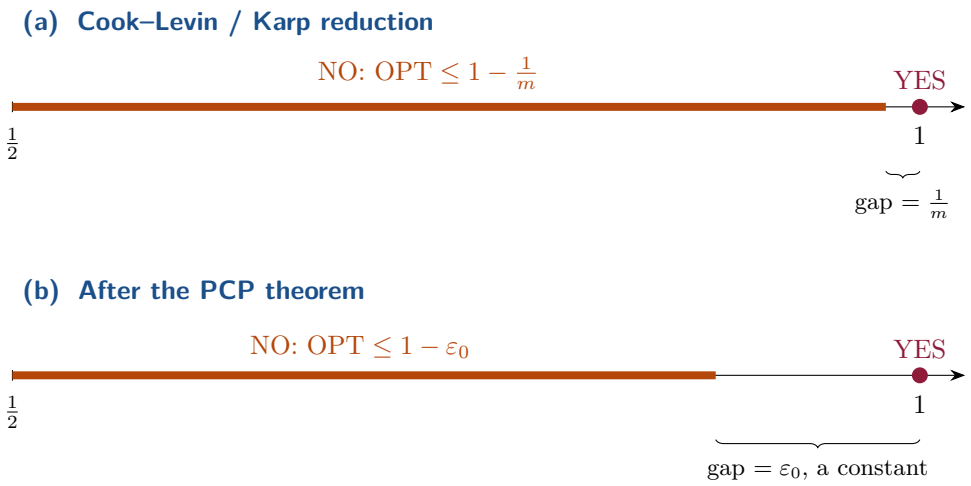


Figure 1.1: Where the optimum $\mathrm{OPT}(\varphi)$ lives for YES and NO instances of a reduction to Max-3SAT (schematic, not to scale). The classical reductions leave a single clause between the two cases; the PCP theorem moves the NO instances away by a constant fraction.

1.5 The PCP theorem: two faces

1.5.1 Proofs that can be spot-checked

A classical NP verifier reads its entire witness. Suppose instead we insist that the verifier be *randomized* and may read only a constant number of bits of a (very long) proof π. Of course it must then sometimes accept wrong proofs; the demand is that it reject every proof of a false statement with probability bounded away from 0.

Definition 1.6 (Probabilistically checkable proofs)**.** A *probabilistic verifier* V with randomness $r(n)$ and query complexity $q(n)$ is a polynomial-time algorithm that, on input x of length n and oracle access to a proof string π, draws a random string $\rho \in \{0,1\}^{r(n)}$, *non-adaptively* computes $q(n)$ positions $i_1(x,\rho), \dots, i_q(x,\rho)$, reads $\pi_{i_1}, \dots, \pi_{i_q}$, and accepts or rejects according to a predicate $P_{x,\rho}\colon \{0,1\}^q \to \{0,1\}$. Let $0 \le s < c \le 1$. A language L belongs to $\mathsf{PCP}_{c,s}[r,q]$ if there is such a verifier with

$$x \in L \implies \exists \pi : \Pr_{\rho}[V^{\pi}(x) \text{ accepts}] \ge c \qquad \text{(completeness)},$$
$$x \notin L \implies \forall \pi : \Pr_{\rho}[V^{\pi}(x) \text{ accepts}] \le s \qquad \text{(soundness)}.$$

We write $\mathsf{PCP}[r,q] = \mathsf{PCP}_{1,1/2}[r,q]$.

Two extreme cases show that the trade-off is not vacuous. With $r = 0$ and $q = n^{O(1)}$ the verifier is deterministic and reads the whole witness: $\mathsf{PCP}[0, \mathrm{poly}(n)] = \mathsf{NP}$. With $q = 0$ the proof is useless, and we get randomized polynomial time. The question is how much *randomness* (which buys coverage of the proof) and how many *queries* are needed. Note that $\mathsf{PCP}[O(\log n), O(1)] \subseteq \mathsf{NP}$ trivially: there are only $2^{O(\log n)} = \mathrm{poly}(n)$ random strings, and together they query only polynomially many positions of π, so a nondeterministic machine can guess these bits and check all random strings. The surprise is the other direction.

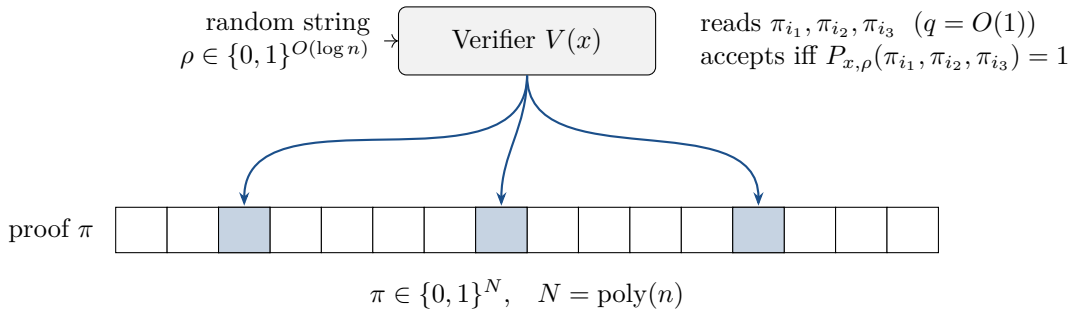


Figure 1.2: A PCP verifier. For each of polynomially many random strings ρ it inspects a constant number of proof bits and applies a local test. A correct proof passes every test; every proof of a false statement fails at least half of them.

Theorem 1.7 (PCP theorem, proof-checking form; Arora–Safra, Arora–Lund–Motwani–Sudan–Szegedy)**.** $\mathsf{NP} = \mathsf{PCP}[O(\log n), O(1)]$.

Theorem 1.8 (PCP theorem, hardness form)**.** *There is a constant $\varepsilon_0 > 0$ such that* $\textsc{Gap}_{1,1-\varepsilon_0}$-3SAT *is* NP*-hard.*

Here $\textsc{Gap}_{1,1-\varepsilon_0}$-3SAT asks to distinguish satisfiable 3-CNF formulas from formulas in which every assignment violates an ε_0-fraction of the clauses. Compare with Figure 1.1: this is precisely the picture in panel (b). The two theorems look very different—one about proofs, one about optimization—but they say the same thing.

Proposition 1.9 (Equivalence of the two forms)**.** *Theorem 1.7 holds if and only if Theorem 1.8 holds.*

Proof. Hardness form ⇒ proof-checking form. Let $L \in \mathsf{NP}$. Compose the Cook–Levin reduction with the reduction witnessing the NP-hardness of $\textsc{Gap}_{1,1-\varepsilon_0}$-3SAT to obtain a polynomial-time map $x \mapsto \varphi_x$ such that $x \in L$ implies that φ_x is satisfiable, and $x \notin L$ implies that every assignment violates at least $\varepsilon_0 m$ clauses, where $m = |\varphi_x|$. The verifier expects as proof a satisfying assignment π of φ_x. It picks a clause uniformly at random ($\log m = O(\log n)$ random bits), reads the (at most) three variables occurring in it, and accepts iff the clause is satisfied. If $x \in L$, an honest proof is accepted with probability 1. If $x \notin L$ then *whatever* π is, it is an assignment and thus violates an ε_0-fraction of the clauses, so V accepts with probability at most $1 - \varepsilon_0$. Now repeat the test $k = \lceil \ln 2/\varepsilon_0 \rceil$ times with independent clauses, accepting iff all k clauses are satisfied. The proof π is fixed once and for all, so the k tests are independent given π and the acceptance probability is at most $(1-\varepsilon_0)^k \le e^{-\varepsilon_0 k} \le \frac{1}{2}$. The verifier uses $k \log m = O(\log n)$ random bits and $3k = O(1)$ queries.

Proof-checking form ⇒ hardness form. Let V be a verifier for 3SAT $\in \mathsf{NP}$ with $r = O(\log n)$ random bits, $q = O(1)$ queries, perfect completeness and soundness $\frac{1}{2}$. (We may assume $q \ge 3$.) Given a formula ψ, we build a 3-CNF φ as follows. Since $2^r = \mathrm{poly}(n)$, we can enumerate every random string ρ and compute the query positions $i_1(\rho), \dots, i_q(\rho)$ and the predicate P_ρ. Introduce one Boolean variable y_i for every position i of the proof. For each ρ we want to enforce $P_\rho(y_{i_1}, \dots, y_{i_q}) = 1$. This constraint is a CNF with at most 2^q clauses of width q: one clause for each $a \in \{0,1\}^q$ with $P_\rho(a) = 0$, namely the clause that is false exactly on the input a. Finally, each clause $(\ell_1 \vee \cdots \vee \ell_q)$

of width q is converted into $q-2$ clauses of width 3 with fresh variables $z_1, \dots, z_{q-3}$:

$$(\ell_1 \vee \ell_2 \vee z_1) \wedge (\neg z_1 \vee \ell_3 \vee z_2) \wedge \cdots \wedge (\neg z_{q-3} \vee \ell_{q-1} \vee \ell_q).$$

If some ℓ_j is true, one can choose the z's to satisfy all $q-2$ clauses; if all ℓ_j are false then z_1 must be true, hence z_2, …, hence z_{q-3}, and the last clause is violated, so *at least one* of the $q-2$ clauses fails for every choice of z. Let φ_ρ be the resulting 3-CNF; it has at most $K := (q-2)2^q$ clauses, and let $\varphi = \bigwedge_\rho \varphi_\rho$, which has $m \le K \cdot 2^r$ clauses.

Completeness. If ψ is satisfiable, there is a proof π accepted for every ρ. Setting $y = \pi$ and choosing the z's suitably satisfies all of φ.

Soundness. If ψ is unsatisfiable, every proof π is rejected for at least half of the random strings. Given any assignment to φ, read off $\pi = y$. For every ρ on which V^π rejects, the assignment violates *at least one* clause of φ_ρ. There are at least $2^r/2$ such ρ, so at least $2^r/2$ clauses are violated out of $m \le K2^r$, i.e. a fraction $\ge 1/(2K)$. Thus $\textsc{Gap}_{1,1-\varepsilon_0}$-3SAT is NP-hard with $\varepsilon_0 = 1/(2K)$, a constant. □

Remark (A dictionary). The proof is our first instance of a translation that we shall use constantly: *a verifier is a constraint satisfaction problem.* Random strings ρ correspond to constraints; positions of the proof correspond to variables; the predicate P_ρ is the constraint; the number q of queries is the arity; soundness s is the bound on the optimum in the NO case. We extend the dictionary to games in Section 1.6 (Table 1.2).

Pitfall: a proof is not a witness

In Definition 1.6 the soundness condition quantifies over *every* string π, including strings that have nothing to do with any computation. A PCP is useful precisely because its verifier cannot be fooled by an adversarially chosen π that is merely *locally plausible*. In the first half of the proof above this is why we can read off an assignment from π and apply the NO-case promise to it.

The PCP theorem is a deep result: the original proofs by Arora and Safra (1998) and Arora et al. (1998) are algebraic (low-degree polynomials and recursive proof composition), and Dinur (2007) later found a purely combinatorial proof by gap amplification on expander graphs. We take Theorem 1.7 for granted in this lecture, return to its proofs in Lectures 3, 4 and 6, and ask here what it gives us.

Remark (Two-sided history). Theorem 1.8 reads as a statement about *approximation*, yet the road to it was paved by results about *interactive proofs*: multi-prover interactive proofs (Ben-Or et al., 1988), NEXP = MIP (Babai et al., 1991), and the discovery by Feige et al. (1996) that a PCP verifier can be turned into a graph whose largest clique encodes the maximum acceptance probability (Exercise 1.4), and hence that PCPs imply hardness of approximating Max-Clique. Section 1.6 follows the thread through the multi-prover systems.

1.6 Two provers, Label Cover, and parallel repetition

The gap in the PCP theorem is a small constant: soundness $1-\varepsilon_0$. To prove hardness results with sharp constants like $\frac{7}{8}$ and $\frac{1}{2}$ we need soundness $s \to 0$, and we need it while keeping the *number of queries* under control. The cleanest setting for this is the following.

1.6.1 Games, labelings, projections

Definition 1.10 (Two-prover one-round game / Label Cover)**.** A *two-prover one-round game* G consists of a bipartite graph (U, W, E) with finite label sets Σ_U, Σ_W and, for each edge $e = (u, w)$, a *constraint* $C_e \subseteq \Sigma_U \times \Sigma_W$. The referee draws an edge $e = (u, w)$ uniformly at random, sends u to Alice and w to Bob, receives answers $a \in \Sigma_U$ and $b \in \Sigma_W$ from the provers (who cannot communicate), and accepts iff $(a, b) \in C_e$. The *value* of G,

$$\mathrm{val}(G) \;=\; \max_{\ell_U : U \to \Sigma_U,\ \ell_W : W \to \Sigma_W} \ \Pr_{e=(u,w)\sim E} \big[(\ell_U(u), \ell_W(w)) \in C_e\big],$$

is the maximum over *labelings*, i.e. over deterministic strategies of the two provers. Finding an optimal labeling is the optimization problem *Label Cover*. The game is a *projection game* if every C_e is the graph of a function $\pi_e \colon \Sigma_U \to \Sigma_W$: $C_e = \{(a, \pi_e(a))\}$. It is a *Unique Game* if moreover $|\Sigma_U| = |\Sigma_W| = k$ and every π_e is a bijection.

Restricting to deterministic strategies loses nothing.

Lemma 1.11 (Determinism suffices)**.** *The maximum acceptance probability over all (randomized, possibly correlated through shared randomness) strategies of Alice and Bob equals* $\mathrm{val}(G)$.

Proof. A randomized strategy with shared random string r is, for each fixed r, a deterministic strategy. Its acceptance probability is the average over r of the acceptance probabilities of the deterministic strategies, hence at most the maximum. □

Lemma 1.12 (Random labeling)**.** *In a projection game,* $\mathrm{val}(G) \geq 1/|\Sigma_W|$. *In particular, for every Unique Game with k labels* $\mathrm{val}(G) \geq 1/k$.

Proof. Fix any labels for the $u \in U$ and give every $w \in W$ an independent uniformly random label. For each edge $e = (u, w)$ the label $\pi_e(\ell_U(u))$ is fixed, so w matches it with probability exactly $1/|\Sigma_W|$. By linearity of expectation some labeling achieves at least that. □

Lemma 1.12 will matter later: no hardness theorem can have soundness δ with fewer than $1/\delta$ labels.

Example 1.13 (The clause–variable game)**.** Let φ be a 3-CNF in which every clause has three distinct variables. Alice plays the *clause prover*, Bob the *variable prover*. The referee picks a uniformly random clause C and a uniformly random variable x in it. Alice receives C and must answer with a satisfying assignment of its three variables (7 possibilities); Bob receives x and answers with a bit. They win iff Bob's bit equals Alice's value for x. This is a projection game with $|\Sigma_U| = 7$ and $|\Sigma_W| = 2$ (Figure 1.3); the projection $\pi_{(C,x)}$ sends an assignment of C to its restriction to x.

Proposition 1.14 (Soundness of the clause–variable game)**.** *If φ is satisfiable then* $\mathrm{val}(G_\varphi) = 1$. *If every assignment violates at least an ε-fraction of the clauses then* $\mathrm{val}(G_\varphi) \leq 1 - \varepsilon/3$.

Proof. If φ is satisfiable, let both provers answer according to a satisfying assignment. Conversely, fix deterministic strategies; Bob's answers define an assignment b to the variables. By assumption b violates at least an ε-fraction of the clauses. Consider a

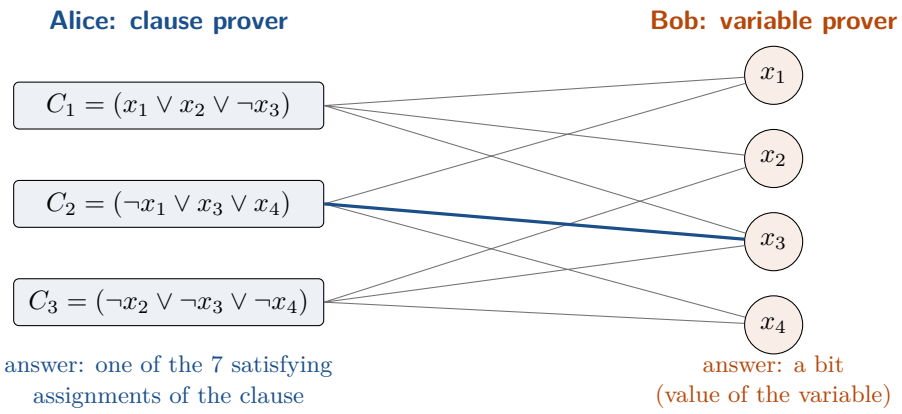


Figure 1.3: The clause–variable game of a 3-CNF. An edge is a pair (clause, variable occurring in it). The highlighted edge (C_2, x_3) is accepted iff Alice's assignment to C_2 and Bob's bit agree on x_3, i.e. iff the projection $\pi_{(C_2,x_3)}$ of Alice's label to the coordinate x_3 equals Bob's label.

PCP verifier	CSP	Two-prover game	Label Cover
random string ρ	constraint	question pair (u, w)	edge e
proof position	variable	(a prover's answer table)	vertex
q queries	arity q	2 provers	arity 2
predicate P_ρ	constraint predicate	referee's test	relation C_e
alphabet $\{0,1\}$	domain	answer set	label set Σ
soundness s	$\mathrm{OPT} \le s$	$\mathrm{val}(G) \le s$	$\mathrm{val} \le s$

Table 1.2: The dictionary. The same object is a verifier, a constraint satisfaction problem, a game, and a Label Cover instance.

clause C violated by b. Alice's answer a_C is a satisfying assignment of C (otherwise she loses outright), so $a_C \neq b|_C$, and therefore a_C disagrees with b on at least one of the three variables of C. The referee asks exactly that variable with probability at least $\frac{1}{3}$, and rejects. Hence the rejection probability is at least $\varepsilon \cdot \frac{1}{3}$. □

Combined with Theorem 1.8, Proposition 1.14 shows that it is NP-hard to distinguish $\mathrm{val}(G) = 1$ from $\mathrm{val}(G) \le 1 - \varepsilon_0/3$ for projection games over alphabets of size 7 and 2. We have now met the "game" column of the dictionary.

1.6.2 Parallel repetition

To drive the soundness from $1 - \varepsilon_0/3$ to a small δ we want to repeat. For a one-oracle verifier, as we saw in the proof of Proposition 1.9, independent repetition multiplies the acceptance probabilities. For a game we are not free to repeat sequentially: that would require several rounds of communication, which we want to avoid (otherwise the provers could coordinate). Instead one repeats *in parallel*.

Definition 1.15 (Parallel repetition)**.** The *k-fold parallel repetition* $G^{\otimes k}$ of a game G: the referee draws k independent edges $e_i = (u_i, w_i)$, sends $(u_1, \dots, u_k)$ to Alice and

$(w_1, \dots, w_k)$ to Bob, receives $a \in \Sigma_U^k$ and $b \in \Sigma_W^k$, and accepts iff $(a_i, b_i) \in C_{e_i}$ for every i.

Alice and Bob can play each coordinate independently with an optimal strategy of G, so $\mathrm{val}(G^{\otimes k}) \geq \mathrm{val}(G)^k$. Is this an equality? It is not, and the failure is dramatic enough to deserve a warning.

Pitfall: parallel repetition is not independent repetition

Let $\ell \geq 3$ be odd. In the *odd cycle game* the questions are the vertices of an ℓ-cycle. With probability $\frac{1}{2}$ the referee sends the *same* vertex to both provers and demands the same color (0 or 1); with probability $\frac{1}{2}$ it sends the two *endpoints of a random edge*, in random order, and demands different colors. The provers are trying to convince the referee that an odd cycle is 2-colorable. Its value is $1 - \frac{1}{2\ell}$ (Exercise 1.6); this is a Unique Game with two labels. Naively, k repetitions should yield value about $(1 - \frac{1}{2\ell})^k \approx e^{-k/2\ell}$, which for $k = \ell^2/100$ is exponentially small in ℓ. In fact Raz (2011) showed that for $k \leq \ell^2$ there is a strategy winning $G^{\otimes k}$ with probability $1 - O(\sqrt{k}/\ell)$, so that even for $k = c\,\ell^2$ with a small constant c the value of $G^{\otimes k}$ is still at least $\frac{1}{2}$ (this is essentially tight by Feige et al., 2007). Exercise 1.7 gives a much smaller example in which already the 2-fold game beats val^2. Examples of this kind go back to Fortnow and Feige around 1990, and for several years it was a famous open problem whether repetition drives the value to 0 at all.

The qualitative fact that repetition *does* drive the value to 0 is the theorem of Raz.

Theorem 1.16 (Parallel repetition; Raz 1998, Holenstein 2009, Rao 2011)**.** *There is an absolute constant $c > 0$ such that for every two-prover one-round game G with $\mathrm{val}(G) \leq 1 - \eta$ and answer sets of size at most σ,*

$$\mathrm{val}(G^{\otimes k}) \;\leq\; (1 - \eta^c)^{\Omega(k/\log \sigma)}.$$

For projection games, Rao showed $\mathrm{val}(G^{\otimes k}) \leq (1 - \Omega(\eta^2))^k$, independently of σ.

Raz's original paper has $c = 32$ (Raz, 1998); Holenstein simplified the proof and improved this to $c = 3$ (Holenstein, 2009). (We prove none of this until Lecture 5.) Putting the PCP theorem, the clause–variable game and parallel repetition together, we obtain the result that all of Lectures 7–16 take as their starting point.

Theorem 1.17 (Label Cover theorem)**.** *For every $\delta > 0$ there is a $\sigma = \mathrm{poly}(1/\delta)$ and a reduction, running in time $n^{O(\log(1/\delta))}$ on formulas of size n, from 3SAT to projection games with label sets of size at most σ, such that*

$$\varphi \text{ satisfiable} \;\Rightarrow\; \mathrm{val}(G_\varphi) = 1, \qquad \varphi \text{ unsatisfiable} \;\Rightarrow\; \mathrm{val}(G_\varphi) \leq \delta.$$

In other words $\textsc{Gap}_{1,\delta}$-$\textsc{LabelCover}_\sigma$ is NP-hard.

Proof. Apply Theorem 1.8 to obtain φ' with the ε_0-gap, then form the clause–variable game $G = G_{\varphi'}$; by Proposition 1.14 we have $\mathrm{val}(G) = 1$ or $\mathrm{val}(G) \leq 1 - \eta$ with $\eta = \varepsilon_0/3$. Take $G^{\otimes k}$ with $k = O(\log(1/\delta)/\eta^2) = O(\log(1/\delta))$. Repetition of a projection game is again a projection game (the projection is $\pi_{e_1} \times \cdots \times \pi_{e_k}$). If $\mathrm{val}(G) = 1$ then $\mathrm{val}(G^{\otimes k}) = 1$. Otherwise Theorem 1.16 gives $\mathrm{val}(G^{\otimes k}) \leq (1 - \Omega(\eta^2))^k \leq \delta$. The new label sets have sizes 7^k and 2^k, which is $\mathrm{poly}(1/\delta)$, and $G^{\otimes k}$ has at most $(3m)^k$ edges. □

Two remarks about the price we paid. First, the alphabet size σ is polynomial in $1/\delta$, and by Lemma 1.12 it *must* be at least $1/\delta$. Second, the projection games we have produced are d-to-1 with large d: each label of Bob has about $\sigma/2^k$ preimages. Both facts will be the stage on which Unique Games enters.

1.7 Unique Games and the Unique Games Conjecture

1.7.1 Definition and a first observation

We now restrict the projection constraints as much as possible. It is convenient (and customary) to define Unique Games on a general graph.

Definition 1.18 (Unique Games)**.** A *Unique Games instance* $\mathcal{U} = (G, [k], \{\pi_e\})$ consists of a graph $G = (V, E)$, a label set $[k]$, and for every edge $e = (u, v)$ a *bijection* $\pi_e \colon [k] \to [k]$. A labeling $\ell \colon V \to [k]$ *satisfies* e if $\pi_e(\ell(u)) = \ell(v)$, and $\mathrm{val}(\mathcal{U}) = \max_\ell \Pr_{e\sim E}[\ell \text{ satisfies } e]$. Since π_e is a bijection, $\pi_e(\ell(u)) = \ell(v) \iff \ell(u) = \pi_e^{-1}(\ell(v))$, so the orientation of an edge is immaterial.

Example 1.19. Max-Cut is a Unique Game with $k = 2$ and $\pi_e =$ the transposition $0 \leftrightarrow 1$ for every edge. More generally, a system of equations $x_u - x_v = c_e \pmod k$ is a Unique Game with $\pi_e(i) = i + c_e$: two-variable linear equations over $\mathbb{Z}_k$. By contrast, Max-3-Cut ($x_u \neq x_v$ with $k = 3$) is *not* a Unique Game: given x_u, the constraint allows two values of x_v, not one (Exercise 1.8).

The word "unique" means that, given the label of one endpoint, the label of the other endpoint is forced. That makes perfect completeness trivial.

Proposition 1.20 (Unique Games with perfect completeness are easy)**.** *Given a Unique Games instance with k labels, one can decide in time $O(k\,(|V|+|E|))$ whether* $\mathrm{val}(\mathcal{U}) = 1$*, and find a perfect labeling if there is one.*

Proof. Treat each connected component separately. Pick a root r and try each of the k labels a for it. Propagate along a BFS tree: if u has label $\ell(u)$ and (u, v) is a tree edge then set $\ell(v) = \pi_e(\ell(u))$ (or $\pi_e^{-1}(\ell(u))$ if the edge is traversed against its orientation). Check all edges. If a perfect labeling ℓ^* exists, then with $a = \ell^*(r)$ the propagation reconstructs ℓ^* by induction along the tree, because bijectivity makes each step forced. So some a passes the check. □

Remark. The argument breaks down for projection games, where the label of u determines that of w but not vice versa: propagating from w to u requires choosing among the d preimages, and the branching is exponential. Indeed, deciding val $= 1$ for projection games is NP-hard (Proposition 1.14). Likewise, as in Section 1.3, if a hard Unique Games gap problem exists, it must have completeness $1 - \varepsilon < 1$: an approximately satisfiable instance may have a bad edge on every propagation path, and the instance has to be built so that no local repair works.

1.7.2 The conjecture

Conjecture 1.21 (Unique Games Conjecture; Khot 2002)**.** *For every $\varepsilon, \delta > 0$ there is a $k = k(\varepsilon, \delta)$ such that* $\mathrm{Gap}_{1-\varepsilon,\delta}$-$\mathrm{UG}_k$ *is* NP*-hard: it is* NP*-hard to distinguish Unique Games instances over k labels with* val $\geq 1 - \varepsilon$ *from those with* val $\leq \delta$*.*

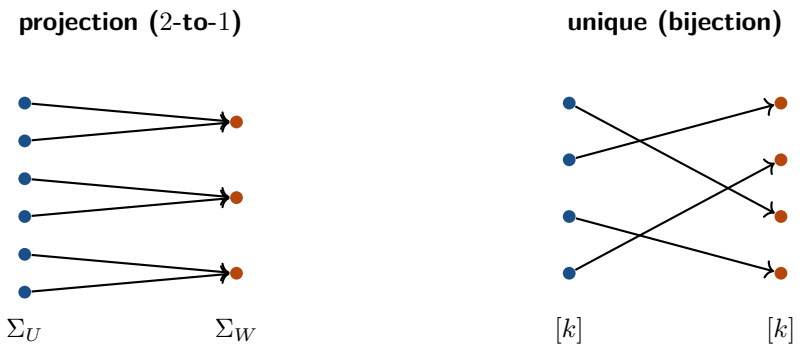


Figure 1.4: A constraint π_e between the label sets of two vertices. In a projection game (left) a label of w has several preimages; in a Unique Game (right) the constraint is a perfect matching.

Every word of this statement carries weight.

(i) *Quantifiers.* First ε and δ are chosen, *then* the alphabet k. The alphabet is a constant but it grows without bound as $\varepsilon, \delta \to 0$. It must: by Lemma 1.12, $k \geq 1/\delta$.

(ii) *A family of problems.* The UGC is not a statement about one problem, but about a family of promise problems parametrised by (ε, δ). (Hence "UG is NP-hard to approximate" is a misleading summary.)

(iii) *Completeness* $1 - \varepsilon$. By Proposition 1.20 we cannot ask for completeness 1.

(iv) *Soundness* $\delta \to 0$. This is the opposite extreme to the PCP theorem, which only gets soundness $1-\varepsilon_0$: the conjecture says that even "almost satisfiable" and "almost completely unsatisfiable" are indistinguishable.

(v) *Reductions.* "NP-hard" means via polynomial-time Karp reductions from 3SAT. Some applications are known to go through under weaker hypotheses, e.g. randomized reductions.

Remark. There are many variants of the conjecture: with bipartite graphs, with linear constraints $\pi_e(i) = i + c_e$ (a special case, believed hard too), with "strong" versions in which soundness is measured on vertex sets rather than edges. Their relationships are surveyed by Khot (2010) and Trevisan (2012). The manuscript discussed in Section 1.8 addresses the usual edge-value form of Conjecture 1.21 with translation constraints on a bipartite graph.

1.7.3 Why Unique Games?

If we can already prove $\text{Gap}_{1,\delta}$-LabelCover is NP-hard for every δ (Theorem 1.17), why would anyone want a new conjecture? Because Label Cover is the wrong tool for exactly the problems in the last two rows of Table 1.1.

The two barriers behind Khot's conjecture

(1) The two-query barrier. Håstad's optimal results for Max-E3-SAT and Max-E3-Lin come from PCPs that read *three* bits. A problem such as Max-Cut, Vertex Cover or Min-2SAT-Deletion has constraints on *two* variables, so a PCP proving its hardness must use only *two* queries—and two-query tests are intrinsically much weaker. Label Cover is the natural two-query starting point, with alphabet as long as we please. **(2) The**

many-to-one barrier. Roughly speaking, a hardness reduction from Label Cover replaces every vertex by a code (the *long code*) and every constraint π_e by a local test between two codewords; the analysis of a cheating prover must *decode* the code tables back into a labeling of the Label Cover instance. When π_e is d-to-1 with large d, the decoding can only recover the label of w up to a list of d candidates for u, and the loss in soundness is too large to reach the thresholds of Table 1.1. When π_e is a bijection there is no such loss. Khot's insight (Khot, 2002) is that one can hope that a *unique* Label Cover is just as hard as general Label Cover, once completeness is relaxed to $1-\varepsilon$, and that this hypothesis—the UGC—would then imply tight hardness for two-variable problems. (We make all of this precise in Lectures 8 and 9.)

1.7.4 Evidence for and against

Nobody knows whether the UGC is true. The state of the evidence, up to this writing, is a mixture of three things.

- *Algorithms.* If val $\geq 1-\varepsilon$, one can find in polynomial time a labeling of value $1-O(\sqrt{\varepsilon \log k})$ (Charikar et al., 2006), and in time $\exp(n^{\varepsilon^{\Omega(1)}})$ a labeling of value $1-\varepsilon^{\Omega(1)}$ (Arora et al., 2015). No algorithm contradicts the conjecture, but the subexponential algorithm shows that Unique Games is not as hard as 3SAT is believed to be (Lecture 12).
- *Integrality gaps.* The semidefinite program that powers the algorithms above cannot refute the conjecture: it has integrality gaps (Khot and Vishnoi, 2015) (Lecture 10).
- *Partial hardness.* The 2-to-2 Games Conjecture, a weakened version of the UGC in which each constraint is 2-to-2 instead of a bijection, is a *theorem* (Khot et al., 2017; Dinur et al., 2018; Khot et al., 2018); see Khot (2019) for an exposition. It already implies NP-hardness of approximating Vertex Cover within $\sqrt{2}-\varepsilon$, the best-known (Lectures 13–15).

1.8 The map of the territory

Figure 1.5 places the main results of this course on a single page; Table 1.3 gives the chronology. Read an arrow as "is used to prove" (or "builds on"). One item on the map needs a health warning.

Status note: the 2026 claim

On 23 September 2026 OpenAI released a manuscript *The Unique Games Theorem* (OpenAI, 2026). Its abstract reads: "We prove the Unique Games Conjecture. For every fixed $\varepsilon, \delta \in (0, 1/2)$, we give a deterministic polynomial-time reduction from 3SAT to Unique Games over a fixed finite alphabet, with completeness at least $1-\varepsilon$ and soundness at most δ."

Status. At the time of writing we know of no independent expert verification of the proof, and no refutation; we have not verified it ourselves. Conjecture 1.21 is therefore treated as open, and every statement labelled "assuming the UGC" in these notes is *conditional*. Lecture 16 compares the manuscript's main theorem with the classical formulations and discusses what a careful verification would have to check. Until then we keep apart results established by the community, results in a complete but unrefereed paper, and claims published by their authors only.

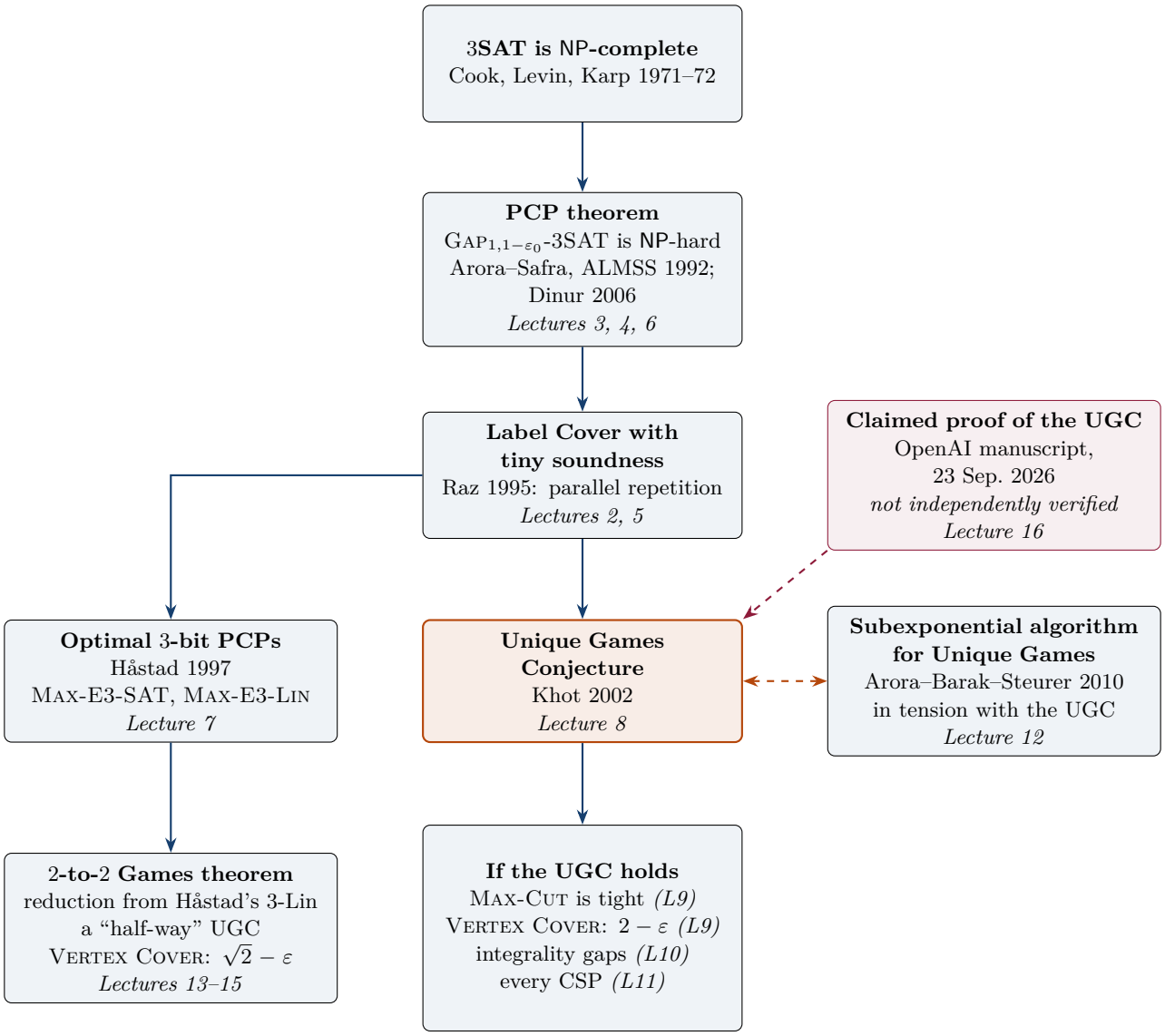


Figure 1.5: The route from NP-completeness to Unique Games. Solid blue arrows are established implications. The orange double arrow records a *tension*: a subexponential algorithm for Unique Games sits uneasily next to a conjecture asserting NP-hardness. The dashed wine-red arrow marks a claim whose status is explained in the box below. *Ln* abbreviates "Lecture n".

Year	Event
1971–72	Cook, Levin, Karp: NP-completeness (Cook, 1971; Levin, 1973; Karp, 1972)
1974	Johnson: systematic study of approximation algorithms (Johnson, 1974)
1988	Ben-Or, Goldwasser, Kilian, Wigderson: multi-prover interactive proofs (Ben-Or et al., 1988)
1990	Babai, Fortnow, Lund: NEXP = MIP (Babai et al., 1991)
1991	Feige, Goldwasser, Lovász, Safra, Szegedy: PCPs imply hardness of Clique (Feige et al., 1996)
1992	Arora–Safra and Arora–Lund–Motwani–Sudan–Szegedy: the PCP theorem (Arora and Safra, 1998; Arora et al., 1998); Feige–Lovász: two-prover one-round proof systems (Feige and Lovász, 1992)
1994	Goemans–Williamson: 0.878-approximation for Max-Cut (Goemans and Williamson, 1995)
1995	Raz: parallel repetition (Raz, 1998); Bellare–Goldreich–Sudan: the long code (Bellare et al., 1998)
1996	Håstad: Clique is hard within $n^{1-\varepsilon}$ (Håstad, 1999)
1997	Håstad: optimal inapproximability for Max-E3-SAT and Max-E3-Lin (Håstad, 2001)
2002	Khot: the Unique Games Conjecture (Khot, 2002)
2003	Khot–Regev: Vertex Cover is hard within $2-\varepsilon$ under the UGC (Khot and Regev, 2008)
2004–05	Khot–Kindler–Mossel–O'Donnell: Max-Cut (Khot et al., 2007); Khot–Vishnoi: integrality gaps (Khot and Vishnoi, 2015)
2006	Dinur: the PCP theorem by gap amplification (Dinur, 2007)
2008	Raghavendra: every CSP (Raghavendra, 2008)
2010	Arora–Barak–Steurer: subexponential algorithm for Unique Games (Arora et al., 2015)
2017–18	Khot–Minzer–Safra, Dinur–Khot–Kindler–Minzer–Safra: 2-to-2 Games (Khot et al., 2017; Dinur et al., 2018; Khot et al., 2018)
2026	OpenAI manuscript claiming the UGC (OpenAI, 2026)

Table 1.3: A timeline of the main results (conference years; journal versions are later). Many important contributions are missing.

Exercises

Exercise 1.1 ([⋆]). Derandomize the three random-assignment algorithms of Example 1.2 with the method of conditional expectations: fix the variables one at a time, always choosing the value that does not decrease the conditional expected number of satisfied constraints. Why is this efficiently computable?

Exercise 1.2 ([⋆]). State and prove the analogue of Lemma 1.4 for minimization problems, where YES instances have OPT $\le c$ and NO instances have OPT $> s$ with $c < s$. What is the inapproximability ratio?

Exercise 1.3 ([⋆]). Show that $\text{Gap}_{1,s}$-Max-E3-Lin is in P for every $s < 1$. (Hence Håstad's theorem cannot have perfect completeness.) Why does the same argument not

apply to $\textsc{Gap}_{1,\frac{7}{8}+\varepsilon}$-Max-E3-SAT?

Exercise 1.4 ([⋆⋆]). Let V be a PCP verifier with r random bits and q queries, and input x. Build a graph H_x whose vertices are the pairs (ρ, a) with $\rho \in \{0,1\}^r$ and $a \in \{0,1\}^q$ satisfying $P_\rho(a) = 1$, and in which (ρ, a) and (ρ', a') are adjacent if $\rho \neq \rho'$ and the two answer vectors do not contradict each other on positions queried by both. Prove that the largest clique of H_x has size $2^r \cdot \max_\pi \Pr_\rho[V^\pi(x) \text{ accepts}]$. Conclude, using the PCP theorem, that it is NP-hard to approximate Max-Clique within some constant factor. (Amplifying this to a polynomial factor needs randomness-efficient repetition; Feige et al., 1996.)

Exercise 1.5 ([⋆]). Let V be a one-oracle verifier with soundness s. Prove that its k-fold independent repetition has soundness s^k. Identify exactly where the argument uses that there is one fixed proof π, and explain why the same argument breaks for parallel repetition of two-prover games.

Exercise 1.6 ([⋆⋆]). Prove that the odd cycle game has value exactly $1 - \frac{1}{2\ell}$. *Hint: for the lower bound, let both provers color the cycle with a single monochromatic edge. For the upper bound, write the strategies as colorings f_A, f_B of the vertices and build a graph on the 2ℓ vertices $(v, A), (v, B)$ with "$\neq$" edges $(u, A)-(v, B)$ for every oriented edge (u, v) of the cycle, and "$=$" chords $(v, A)-(v, B)$. The "$\neq$" edges form a single even cycle of length 2ℓ, and the two endpoints of any chord are at odd distance ℓ along it. So each chord together with either arc of length ℓ forms a cycle (2ℓ of them in total) that no coloring can satisfy completely. Every "$\neq$" edge has probability $\frac{1}{4\ell}$ and lies in ℓ of these cycles, every chord has probability $\frac{1}{2\ell}$ and lies in 2 of them; give each cycle weight $\frac{1}{4\ell^2}$.*

Exercise 1.7 ([⋆⋆]). *Repetition can beat the product.* Consider the following game G. Alice and Bob receive independent uniform bits x and y and answer with bits a and b. They win iff $(x, y) = (0, 0)$ and $a \neq b$, or $(x, y) \in \{(0, 1), (1, 0)\}$ and $a = b = 0$. (For $(x, y) = (1, 1)$ they always lose.) Show that $\mathrm{val}(G) = \frac{1}{2}$ but $\mathrm{val}(G^{\otimes 2}) \geq \frac{3}{8} > \mathrm{val}(G)^2 = \frac{1}{4}$. *Hint: in $G^{\otimes 2}$ let both players use the same strategy, answering the pair (a_1, a_2) as a function of their input pair: $00 \mapsto 00$, $01 \mapsto 10$, $10 \mapsto 01$, $11 \mapsto 00$.*

Exercise 1.8 ([⋆]). (a) Prove that Max-3-Cut is not a Unique Game. (b) Prove that a Unique Game with val $= 1$ and a Label Cover instance with val $= 1$ can differ in complexity by explaining, using results of this lecture, why the first is in P and the second is NP-hard.

Exercise 1.9 ([⋆⋆]). Which of the following statements implies the UGC, which is implied by it, and which is neither? (a) For every $\varepsilon, \delta > 0$ and *every* $k \geq k_0(\varepsilon, \delta)$, $\textsc{Gap}_{1-\varepsilon,\delta}$-$\mathrm{UG}_k$ is NP-hard. (b) There exist $\varepsilon, \delta > 0$ such that for all k, $\textsc{Gap}_{1-\varepsilon,\delta}$-$\mathrm{UG}_k$ is NP-hard. (c) For every ε there is $\delta > 0$ and k such that $\textsc{Gap}_{1-\varepsilon,\delta}$-$\mathrm{UG}_k$ is NP-hard.

Exercise 1.10 ([⋆⋆]). *Reading.* Read the definition of Unique Games in Section 2 of Khot (2010) and the statement of Theorem 1.1 in OpenAI (2026). List all differences in the formulation (type of graph, type of constraint, alphabet, role of determinism, order of quantifiers). For each difference, decide whether it is a cosmetic restatement, a strengthening, or a weakening of the classical conjecture.

Notes and further reading

Where this lecture comes from. The equivalence of the two PCP formulations is standard; see Lecture 1 of Harsha (2007), or Lecture 1 and Lecture 3 of Trevisan (2006), and Chapter 11 of Arora and Barak (2009). That the formulas produced by Cook's theorem are "fragile"—they are almost satisfiable even for NO instances, so a gap cannot be created by such a reduction—is the starting point of Trevisan's first lecture; the gap-problem formalism of Section 1.3 follows Harsha (2007). The DIMACS tutorial notes (Harsha and Charikar, 2010) are an excellent companion for Lectures 1–9. For the Unique Games Conjecture, the survey of Khot (2010) and the Bulletin article of Trevisan (2012) are the primary references.

On attributions. The $n^{1-\varepsilon}$ hardness of Max-Clique is due to Håstad (1999), and was proved under the assumption $\mathsf{NP} \not\subseteq \mathsf{ZPP}$; the assumption was removed by Zuckerman (2007). It is a different paper from the one containing the $\frac{7}{8}$ and $\frac{1}{2}$ results (Håstad, 2001). The first examples of games whose parallel repetition does not behave like a product were due to Fortnow and to Feige; the odd cycle game and its tilings of space are studied by Feige et al. (2007) and Raz (2011).

What is not covered. We have not defined the *long code*, nor discussed the SDP relaxations of Unique Games, nor the connection between the UGC and the *Small-Set Expansion* problem. They enter in later lectures. Lecture 2 develops the formalism of games and Label Cover that we have only sketched.

Bibliography

S. Arora and B. Barak. *Computational Complexity: A Modern Approach*. Cambridge University Press, 2009.

S. Arora and S. Safra. Probabilistic checking of proofs: a new characterization of NP. *Journal of the ACM*, 45(1):70–122, 1998.

S. Arora, C. Lund, R. Motwani, M. Sudan, and M. Szegedy. Proof verification and the hardness of approximation problems. *Journal of the ACM*, 45(3):501–555, 1998.

S. Arora, B. Barak, and D. Steurer. Subexponential algorithms for unique games and related problems. *Journal of the ACM*, 62(5):Article 42, 2015.

L. Babai, L. Fortnow, and C. Lund. Non-deterministic exponential time has two-prover interactive protocols. *Computational Complexity*, 1(1):3–40, 1991.

M. Bellare, O. Goldreich, and M. Sudan. Free bits, PCPs, and nonapproximability—towards tight results. *SIAM Journal on Computing*, 27(3):804–915, 1998.

M. Ben-Or, S. Goldwasser, J. Kilian, and A. Wigderson. Multi-prover interactive proofs: how to remove intractability assumptions. In *Proceedings of the 20th Annual ACM Symposium on Theory of Computing (STOC)*, pages 113–131, 1988.

M. Charikar, K. Makarychev, and Y. Makarychev. Near-optimal algorithms for unique games. In *Proceedings of the 38th Annual ACM Symposium on Theory of Computing (STOC)*, pages 205–214, 2006.

S. A. Cook. The complexity of theorem-proving procedures. In *Proceedings of the 3rd Annual ACM Symposium on Theory of Computing (STOC)*, pages 151–158, 1971.

I. Dinur. The PCP theorem by gap amplification. *Journal of the ACM*, 54(3):Article 12, 2007.

I. Dinur, S. Khot, G. Kindler, D. Minzer, and M. Safra. Towards a proof of the 2-to-1 games conjecture? In *Proceedings of the 50th Annual ACM Symposium on Theory of Computing (STOC)*, pages 376–389, 2018.

U. Feige and L. Lovász. Two-prover one-round proof systems: their power and their problems. In *Proceedings of the 24th Annual ACM Symposium on Theory of Computing (STOC)*, pages 733–744, 1992.

U. Feige, S. Goldwasser, L. Lovász, S. Safra, and M. Szegedy. Interactive proofs and the hardness of approximating cliques. *Journal of the ACM*, 43(2):268–292, 1996.

U. Feige, G. Kindler, and R. O'Donnell. Understanding parallel repetition requires understanding foams. In *Proceedings of the 22nd IEEE Conference on Computational Complexity (CCC)*, 2007.

M. X. Goemans and D. P. Williamson. Improved approximation algorithms for maximum cut and satisfiability problems using semidefinite programming. *Journal of the ACM*, 42(6):1115–1145, 1995.

P. Harsha. CMSC 39600: PCPs, codes and inapproximability, lecture 1. Lecture notes, scribe J. A. Grochow, University of Chicago, 2007.

P. Harsha and M. Charikar. Limits of approximation algorithms: PCPs and unique

games (DIMACS tutorial lecture notes). arXiv:1002.3864, 2010.

J. Håstad. Clique is hard to approximate within $n^{1-\epsilon}$. *Acta Mathematica*, 182(1):105–142, 1999.

J. Håstad. Some optimal inapproximability results. *Journal of the ACM*, 48(4):798–859, 2001.

T. Holenstein. Parallel repetition: simplification and the no-signaling case. *Theory of Computing*, 5:141–172, 2009.

D. S. Johnson. Approximation algorithms for combinatorial problems. *Journal of Computer and System Sciences*, 9(3):256–278, 1974.

R. M. Karp. Reducibility among combinatorial problems. In R. E. Miller and J. W. Thatcher, editors, *Complexity of Computer Computations*, pages 85–103. Plenum Press, 1972.

S. Khot. On the power of unique 2-prover 1-round games. In *Proceedings of the 34th Annual ACM Symposium on Theory of Computing (STOC)*, pages 767–775, 2002.

S. Khot. On the unique games conjecture (invited survey). In *Proceedings of the 25th IEEE Conference on Computational Complexity (CCC)*, pages 99–121, 2010.

S. Khot. On the proof of the 2-to-2 games conjecture. Expository article, 2019.

S. Khot and O. Regev. Vertex cover might be hard to approximate to within $2-\epsilon$. *Journal of Computer and System Sciences*, 74(3):335–349, 2008.

S. Khot and N. K. Vishnoi. The unique games conjecture, integrality gap for cut problems and embeddability of negative-type metrics into ℓ_1. *Journal of the ACM*, 62(1):Article 8, 2015.

S. Khot, G. Kindler, E. Mossel, and R. O'Donnell. Optimal inapproximability results for MAX-CUT and other 2-variable CSPs? *SIAM Journal on Computing*, 37(1):319–357, 2007.

S. Khot, D. Minzer, and M. Safra. On independent sets, 2-to-2 games, and Grassmann graphs. In *Proceedings of the 49th Annual ACM Symposium on Theory of Computing (STOC)*, pages 576–589, 2017. Journal version: Theory of Computing 21(10), 2025.

S. Khot, D. Minzer, and M. Safra. Pseudorandom sets in Grassmann graph have near-perfect expansion. In *Proceedings of the 59th IEEE Symposium on Foundations of Computer Science (FOCS)*, pages 592–601, 2018.

L. A. Levin. Universal sequential search problems. *Problems of Information Transmission*, 9(3):115–116, 1973.

E. Mossel, R. O'Donnell, and K. Oleszkiewicz. Noise stability of functions with low influences: invariance and optimality. *Annals of Mathematics*, 171(1):295–341, 2010.

OpenAI. The unique games theorem. Manuscript, dated September 23, 2026; not independently verified at the time of writing, 2026.

P. Raghavendra. Optimal algorithms and inapproximability results for every CSP? In *Proceedings of the 40th Annual ACM Symposium on Theory of Computing (STOC)*, pages 245–254, 2008.

R. Raz. A parallel repetition theorem. *SIAM Journal on Computing*, 27(3):763–803, 1998.

R. Raz. A counterexample to strong parallel repetition. *SIAM Journal on Computing*,

40(3):771–777, 2011.

L. Trevisan. CS294: PCP and hardness of approximation, lecture notes (lectures 1 and 3). University of California, Berkeley, 2006.

L. Trevisan. On Khot's unique games conjecture. *Bulletin of the American Mathematical Society*, 49(1):91–111, 2012.

L. Trevisan, G. B. Sorkin, M. Sudan, and D. P. Williamson. Gadgets, approximation, and linear programming. *SIAM Journal on Computing*, 29(6):2074–2097, 2000.

D. Zuckerman. Linear degree extractors and the inapproximability of max clique and chromatic number. *Theory of Computing*, 3:103–128, 2007.